Section A – IPv4 Addressing, Subnet Masks, and Subnet Calculations

Question 1. IPv4 Address Structure and Network Identification

An industrial controller has been assigned the IPv4 address 192.168.14.250 with a subnet mask of 255.255.255.0. The network engineer must determine how the address is represented and which portion identifies the local network. Explain the structure of an IPv4 address, convert the assigned address into binary notation, identify its network and host portions, and discuss how this information is used for routing and local communication.

Good Answer:

1. An IPv4 address contains 32 binary bits organized into four eight-bit octets, each represented by a decimal value from 0 to 255.
2. The binary representation of 192.168.14.250 is 11000000.10101000.00001110.11111010.
3. The subnet mask 255.255.255.0, or /24, identifies the first 24 bits as the network portion and the final eight bits as the host portion.
4. The network address is 192.168.14.0, and the host portion has the decimal value 250.
5. The network portion determines the subnet used for routing decisions, while the host portion identifies the controller's interface within that subnet.

Lecture reference: Section 4.1 – IPv4 Addressing, Network ID, Host ID, and Subnet Masks, Slides 3, 7–10.

Question 2. Binary AND Operation and Partial-Octet Subnetting

A PLC is configured with the IPv4 address 172.16.67.5 and subnet mask 255.255.192.0. The network engineer must determine the subnet to which the PLC belongs and verify the range of addresses that can be assigned to other devices on the same subnet. Using the logical AND operation and the subnet increment method, determine the network address, broadcast address, usable host range, and maximum number of traditional usable host addresses.

Good Answer:

1. The subnet mask 255.255.192.0 corresponds to /18, meaning 18 bits identify the network and 14 bits remain for host addressing.
2. The third octet of the mask is 192, giving a subnet increment of (256 – 192 = 64), so subnet blocks begin at 0, 64, 128, and 192.
3. Since the third octet of the IP address is 67, the address belongs to the 64 - 127 block, giving the network address 172.16.64.0.
4. The broadcast address is 172.16.127.255, and the usable host range is 172.16.64.1 through 172.16.127.254.
5. With 14 host bits, the subnet contains (2^14 = 16,384) total addresses and (16,384 – 2 = 16,382) traditional usable host addresses.

Lecture reference: Section 4.1 – Finding the Network Address Using AND and Partial-Octet Subnetting, Slides 9 and 11.

Question 3. Subnet Increment and Network Boundary Analysis

An industrial Ethernet device has the address 172.20.11.55/22. During troubleshooting, an engineer needs to determine whether other devices belong to the same IP subnet. Calculate the subnet mask, network address, broadcast address, and valid host range for this device. Explain how the subnet increment is used to identify the correct network boundary.

Good Answer:

1. A /22 prefix contains 22 network bits and corresponds to the subnet mask 255.255.252.0.
2. The third octet has a mask value of 252, giving a subnet increment of (256-252=4).
3. Since the third octet is 11, the address belongs to the block from 8 through 11, giving the network address 172.20.8.0.
4. The broadcast address is 172.20.11.255, and the valid host range is 172.20.8.1 through 172.20.11.254.
5. The /22 network has 10 host bits, providing (2^10-2=1,022) traditional usable host addresses, and devices within this subnet can communicate locally when the Layer 2 network permits it.

Lecture reference: Section 4.1 – CIDR and Finding the Subnet Increment, Slides 18 and 31.

Question 4. Subnet Design Using Borrowed Host Bits

An industrial plant has been assigned the network 172.20.0.0/16. The engineering department wants to divide this network into at least 20 equal-size subnets to support different production areas. Determine the number of bits that must be borrowed, the new subnet prefix and subnet mask, the number of available subnets, and the maximum traditional usable hosts per subnet. Explain the engineering tradeoff involved.

Good Answer:

1. The number of borrowed host bits must satisfy ($2^n \geq 20$), so five bits are required because ($2^5=32$).
2. Borrowing five bits from the original /16 network produces a new prefix length of /21.
3. The /21 subnet mask is 255.255.248.0, providing 32 possible equal-size subnets within the original address block.
4. Each subnet has (32-21=11) host bits, giving ($2^{11}-2=2,046$) traditional usable host addresses.
5. Borrowing host bits increases the number of available subnets but reduces the number of host addresses within each subnet, requiring the engineer to balance network segmentation against future equipment expansion.

Lecture reference: Section 4.1 – Calculating a New Subnet, Slide 30.

Question 5. Variable Length Subnet Masking (VLSM) for Industrial Network Design

A manufacturing company has been allocated the IPv4 address block 192.168.0.0/23. Three production departments require different numbers of networked devices: Department A requires 200 hosts, Department B requires 50 hosts, and Department C requires 20 hosts. Design an efficient addressing plan using Variable Length Subnet Masking (VLSM). Determine the subnet allocation for each department, its usable host capacity, and explain why VLSM is preferable to assigning equal-size subnets.

Good Answer:

1. Department A requires at least 200 host addresses, so 192.168.0.0/24 is assigned, providing 254 traditional usable host addresses.
2. Department B requires 50 host addresses, so 192.168.1.0/26 is assigned, providing 62 usable addresses.
3. Department C requires 20 host addresses, so 192.168.1.64/27 is assigned, providing 30 usable addresses.
4. The three allocations do not overlap and fit inside the original 192.168.0.0/23 address block, leaving unallocated address space available for future use.
5. VLSM improves address utilization by allocating subnet sizes according to actual device requirements rather than assigning the same number of addresses to every department.

Lecture reference: Section 4.1 – Advantages of CIDR and VLSM, Slide 19.

Section B – IPv4 Address Allocation, Broadcast Domains, and Address Translation

Question 6. Broadcast Domains and Industrial Network Segmentation

A manufacturing plant has installed hundreds of PLCs, HMIs, industrial computers, and smart sensors on a single large Ethernet network. As more devices are added, network administrators observe increased broadcast traffic and greater difficulty isolating communication faults. Explain how broadcast domains affect industrial network performance and analyze how subnetting, VLANs, and Layer 3 routing can improve network organization and communication management.

Good Answer:

1. A broadcast domain consists of devices that receive Layer 2 broadcast frames transmitted within the same network segment or VLAN.
2. Industrial devices may generate broadcast traffic through services such as ARP requests, DHCP discovery, and device-discovery protocols.
3. A large broadcast domain increases the number of devices receiving and examining broadcast frames, creating additional processing and network overhead.
4. Dividing the network into smaller IP subnets implemented with separate VLANs can limit broadcast traffic to the appropriate equipment groups.
5. Routers and Layer 3 switches provide communication between the separated networks while enabling better traffic management, fault isolation, and application of security policies.

Lecture reference: Section 4.1 – Broadcast Domains and Subnet Size, Slides 19–20 and 29.

Question 7. Static vs. Dynamic IP Address Allocation in Industrial Automation

A manufacturing facility contains PLCs, HMIs, production servers, maintenance laptops, and engineering workstations. The network administrator must decide which devices should use static IP addresses and which should receive addresses dynamically through DHCP. Compare static and dynamic IP addressing and propose an appropriate addressing strategy for the facility. Explain how DHCP parameters affect communication reliability and network administration.

Good Answer:

1. Static IP addresses are configured to remain consistent and are suitable for devices such as PLCs, HMIs, and servers when predictable addressing is required.
2. Dynamic addressing uses DHCP to automatically assign network configuration information, reducing the need for manual configuration of many client devices.
3. DHCP can provide an IP address, subnet mask, default gateway, and DNS server information to a network device.
4. Maintenance laptops and general-purpose engineering workstations may use DHCP, while fixed control equipment may use manually assigned addresses or carefully managed DHCP reservations.
5. A reliable industrial addressing plan must prevent duplicate IP addresses, preserve connectivity to important equipment, and ensure that all devices receive correct subnet and gateway settings.

Lecture reference: Section 4.1 – Static, Dynamic, and Private Addresses, Slide 5; Rules for Assigning IPv4 Addresses, Slides 35–36.

Question 8. Public and Private IPv4 Address Allocation

A manufacturing organization is developing a network that includes multiple production plants, corporate offices, and remote monitoring facilities. Most industrial devices require internal communication, but selected systems must also exchange information with external services. Explain how public IPv4 addresses are allocated globally, identify the private IPv4 ranges available for internal networks, and discuss why private addressing is commonly used in industrial automation.

Good Answer:

1. The Internet Assigned Numbers Authority (IANA) coordinates the global IP address pool and allocates address blocks to Regional Internet Registries (RIRs).
2. Regional Internet Registries distribute address resources to Internet service providers and other organizations, while ISPs assign public addresses to their customers.
3. The three private IPv4 address ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.
4. Private IPv4 addresses can be reused by different organizations because they are not globally routed through the public Internet.
5. Industrial organizations commonly use private addresses for PLCs, HMIs, and internal equipment networks, while network boundary devices can provide controlled external connectivity when required.

Lecture reference: Section 4.1 – IPv4 Address Allocation Hierarchy and Private IP Addresses, Slides 4 and 21.

Question 9. Static NAT, Dynamic NAT, and PAT Selection

An industrial company operates several internal production servers, engineering workstations, and monitoring computers using private IPv4 addresses. Some servers must be reachable through consistent external addresses, while many engineering computers need occasional Internet access. Compare Static NAT, Dynamic NAT, and Port Address Translation (PAT), and recommend appropriate applications for each method. Discuss their differences in address mapping and public IPv4 address consumption.

Good Answer:

1. Static NAT creates a permanent one-to-one mapping between a private IPv4 address and a public IPv4 address.
2. Static NAT is suitable for an industrial gateway or server requiring a consistent external address mapping.
3. Dynamic NAT assigns available public addresses from a pool when internal devices require external communication, so mappings may change over time.
4. PAT allows multiple private devices to share one or a few public addresses by translating TCP or UDP port numbers in addition to IP addresses.
5. PAT generally provides the most efficient public IPv4 address utilization for large numbers of outgoing connections, while static NAT is useful where fixed address mapping is required.

Lecture reference: Section 4.1 – NAT, Static/Dynamic NAT, and PAT, Slides 22–26.

Question 10. APIPA Addressing and DHCP Communication Failure

An engineering laptop is configured to obtain its IPv4 address automatically from a DHCP server. After being connected to a factory Ethernet network, it receives the address 169.254.88.200 with subnet mask 255.255.0.0. The laptop cannot access a production server on another subnet. Explain why the laptop received this address, identify the addressing mechanism involved, and describe how the engineer should troubleshoot the connectivity problem.

Good Answer:

1. The address 169.254.88.200 belongs to the IPv4 link-local range 169.254.0.0/16, commonly associated with Automatic Private IP Addressing (APIPA).
2. Link-local addressing is typically used when a device configured for DHCP cannot obtain an address because no DHCP server responds.
3. APIPA allows devices with compatible link-local addresses on the same local network to communicate without requiring manual IPv4 configuration.
4. Link-local addresses are not normally routed to other networks and therefore do not provide ordinary access to remote production servers.
5. The engineer should verify the DHCP server, Ethernet connection, network configuration, and DHCP availability, then request a renewed address after correcting the problem.

Lecture reference: Section 4.1 – IPv4 Link-Local Addressing and Special IPv4 Address Ranges, Slides 27–28.

Question 11. Valid Host Addresses and Special IPv4 Address Ranges

During commissioning, an engineer discovers several unusual IPv4 configurations: one device uses 10.1.0.0/16, another uses 10.1.255.255/16, a diagnostic utility uses 127.0.0.1, and a network message is addressed to 255.255.255.255. Evaluate the meaning of these addresses and explain why network, broadcast, loopback, and limited-broadcast addresses cannot be treated as ordinary host addresses.

Good Answer:

1. For the subnet 10.1.0.0/16, the address 10.1.0.0 is the network address because all host bits are zero.
2. The address 10.1.255.255 is the subnet's directed broadcast address because all 16 host bits are set to one.
3. The traditional usable host range for this subnet is 10.1.0.1 through 10.1.255.254, and each assigned host address must be unique.
4. The address 127.0.0.1 belongs to the loopback range and is used to test communication with the local computer rather than an external network device.
5. The address 255.255.255.255 represents a limited IPv4 broadcast intended for the local network segment and is not normally forwarded by routers.

Lecture reference: Section 4.1 – Special and Private IPv4 Address Ranges and Valid Host Addresses, Slides 28 and 36.

Section C – IPv4 Routing, Default Gateways, and Troubleshooting

Question 12. Default Gateway Configuration and Communication Failure

A PLC has the IPv4 address 10.1.0.100/16, but its default gateway is configured as 10.2.0.1. The PLC can communicate with some local devices but cannot access a production server located on another subnet. Analyze the IP addressing configuration, identify why the default gateway is inappropriate, and explain how the gateway should be configured to support communication between different networks.

Good Answer:

1. The PLC address 10.1.0.100/16 belongs to the network 10.1.0.0/16, with a subnet mask of 255.255.0.0.
2. The configured default gateway 10.2.0.1 belongs to a different /16 network and is therefore not directly reachable through the PLC's local subnet under normal routing conditions.
3. A default gateway should normally be a router interface connected to the same local subnet as the PLC.
4. A suitable gateway could be 10.1.0.1, provided that this address is assigned to an available router interface with appropriate routes to the destination network.
5. The PLC sends packets directly to local-subnet destinations but forwards packets for other networks to the configured default gateway, making correct gateway addressing essential for inter-network communication.

Lecture reference: Section 4.1 – Default Gateway and Communication Between Networks, Slides 37–38.

Question 13. Route Summarization and Supernetting

A manufacturing facility has eight production networks, ranging from 172.16.200.0/24 through 172.16.207.0/24. All eight networks can be reached through the same internal router at 172.16.208.250. Explain how route summarization can reduce routing-table complexity. Calculate the summary network address and mask, identify the covered address range, and write the appropriate Windows static route command.

Good Answer:

1. The eight networks have consecutive third-octet values from 200 through 207, allowing them to be represented by one larger CIDR prefix.
2. The common high-order bits produce a /21 summary prefix, corresponding to the subnet mask 255.255.248.0.
3. The resulting summary network is 172.16.200.0/21, covering addresses from 172.16.200.0 through 172.16.207.255.
4. The Windows command is route add 172.16.200.0 mask 255.255.248.0 172.16.208.250, directing packets for all eight networks to the same internal next-hop router.

   It tells Windows: “Whenever you need to communicate with any device in the IP address range 172.16.200.0 through 172.16.207.255, send the packets to the router at 172.16.208.250.”
5. Route summarization reduces routing-table entries and simplifies network administration without requiring the eight original /24 networks to be combined into one physical LAN.

Lecture reference: Section 4.1 – Supernetting and Route Summarization, Slides 33–34 and 47.

Question 14. Routing Table Selection and Route Metrics

An engineering laptop has both wired Ethernet and Wi-Fi connections. The operating system contains two default routes: one through Ethernet with metric 10 and another through Wi-Fi with metric 20. The routing table also includes a specific route to an industrial subnet with a /24 prefix. Explain how the operating system chooses an appropriate route and analyze the roles of destination prefixes, route metrics, directly connected routes, and default gateways.

Good Answer:

1. The operating system examines the destination IP address and searches its routing table for matching network routes.
2. The most specific matching prefix is normally preferred, so a matching /24 route takes priority over a less specific /16 or default /0 route.
3. When otherwise comparable routes exist, route metrics help determine which path is preferred, with the lower metric normally receiving priority.
4. For the two default routes in this example, Ethernet with metric 10 is preferred over Wi-Fi with metric 20 while the Ethernet path is available.
5. If the preferred path becomes unavailable, the operating system may select another valid route, while directly connected destinations normally use an on-link route without requiring a default gateway.

Lecture reference: Section 4.1 – Multiple Default Gateways, Route Metrics, and the Windows Route Command, Slides 43–44.

Question 15. Systematic IPv4 Troubleshooting Using Windows Commands

An engineering workstation cannot communicate reliably with a PLC on an industrial Ethernet network. The network administrator suspects an incorrect IP configuration, outdated address-resolution information, or a routing problem. Develop a systematic troubleshooting procedure using ipconfig, ping, arp, route, and netsh. Explain what each command contributes to diagnosing and correcting IPv4 communication problems.

Good Answer:

1. The engineer begins with ipconfig /all to inspect the workstation's IP address, subnet mask, default gateway, DNS servers, MAC address, and DHCP configuration.
2. The ping command can test whether the PLC or gateway responds to ICMP Echo Requests and help identify latency or intermittent packet loss, although a failed ping does not necessarily prove the device is offline.
3. The command arp -a displays cached IPv4-to-MAC address mappings, helping identify possible address-resolution problems or incorrect cached information.
4. The command route print displays the Windows routing table so the engineer can verify destination prefixes, gateways, interfaces, and route metrics.
5. The netsh utility can inspect or configure network-interface settings, while commands such as ipconfig /renew and ipconfig /flushdns may help resolve DHCP or DNS-related problems when appropriate.

Lecture reference: Section 4.1 – Windows Route, Netsh, Ipconfig, Ping, and ARP Commands, Slides 44 and 48–51.

Section D – IPv6 Addressing, Configuration, and Network Design

Question 16. IPv6 Address Representation and Prefix Identification

An industrial server is assigned the IPv6 address: 2001:0db8:1234:0001:0000:0000:0000:0010/64. Convert the address to its shortest valid notation, identify the network prefix and interface identifier, and explain the rules used to simplify IPv6 addresses. Discuss how the /64 prefix is interpreted in a typical IPv6 LAN.

Good Answer:

1. An IPv6 address consists of 128 bits divided into eight 16-bit groups, each normally represented by four hexadecimal digits.
2. Leading zeros can be removed from each group, so 0db8 becomes db8, 0001 becomes 1, and 0010 becomes 10.
3. Consecutive all-zero groups can be compressed using ::, producing the shortened address 2001:db8:1234:1::10/64.
4. The /64 prefix identifies the first 64 bits as the subnet prefix, represented by 2001:db8:1234:1::/64, while the remaining 64 bits form the interface identifier.
5. The :: compression symbol can appear only once in an IPv6 address to avoid ambiguity, and the 2001:db8::/32 range used in this example is reserved for documentation and training.

Lecture reference: Section 4.2 – IPv6 Address Structure and Special-Purpose Addresses, Slides 54 and 59.

Question 17. IPv6 Interface Identifier Generation Using Modified EUI-64

An industrial Ethernet device has the MAC address 00-0C-29-7C-F9-C4. The network engineer is studying the Modified EUI-64 method for generating an IPv6 interface identifier from the device's MAC address. Explain the Modified EUI-64 conversion process, calculate the resulting 64-bit interface identifier, and discuss why modern systems may prefer other methods of generating IPv6 interface identifiers.

Good Answer:

1. A conventional Ethernet MAC address contains 48 bits, while the interface identifier in a typical IPv6 /64 subnet contains 64 bits.
2. Modified EUI-64 converts the 48-bit MAC address into a 64-bit identifier by inserting the hexadecimal bytes FFFE between the first three and last three bytes of the MAC address.
3. The method also inverts the Universal/Local bit in the first byte, changing the initial byte 00 to 02 in this example.
4. The resulting interface identifier is 020c:29ff:fe7c:f9c4.
5. Modern operating systems often use stable, random, or temporary privacy-based interface identifiers instead of directly embedding MAC-derived information, helping reduce device tracking.

Lecture reference: Section 4.2 – IPv6 Interface Identifier, Slide 55.

Question 18. IPv6 Subnetting and Hierarchical Network Design

An industrial organization receives the IPv6 prefix 2001:db8:a00::/48 for a network design exercise. The organization plans to assign separate /64 subnets to production lines, PLC networks, engineering workstations, and industrial monitoring systems. Determine how many /64 subnets can be created from the allocated /48 prefix, identify the bits used for subnet addressing, and provide examples of subnet allocations. Explain why IPv6 subnetting focuses more on organization and routing than address conservation.

Good Answer:

1. The organization has a /48 IPv6 prefix, meaning the first 48 bits identify its allocated address block.
2. Dividing the prefix into /64 subnets provides (64-48=16) additional bits for subnet identification.
3. The number of available /64 subnets is (2^16=65,536).
4. Example allocations include 2001:db8:a00:0::/64 for PLCs, 2001:db8:a00:1::/64 for HMIs, and 2001:db8:a00:2::/64 for engineering workstations.
5. IPv6 subnetting primarily supports hierarchical network organization, segmentation, and efficient routing rather than conserving individual addresses because the IPv6 address space is extremely large.

Lecture reference: Section 4.2 – IPv6 Subnetting, Slide 60. The 2001:db8 prefix is reserved for documentation and is used here for instructional purposes.

Question 19. IPv6 Unicast, Multicast, and Anycast Communication

An industrial monitoring system must support three communication requirements: sending commands to an individual PLC, distributing information to multiple networked devices, and allowing clients to access one of several available monitoring servers. Compare IPv6 unicast, multicast, and anycast communication and recommend the appropriate method for each requirement. Explain how IPv6 multicast differs from traditional IPv4 broadcast communication.

Good Answer:

1. IPv6 unicast provides one-to-one communication and is appropriate when the monitoring system must communicate with a specific PLC interface.
2. IPv6 multicast sends a packet to a group of subscribed interfaces and is suitable when several industrial devices need to receive the same information.
3. IPv6 anycast allows multiple devices to share an address, with routing directing traffic to one selected instance according to the routing system's path selection.
4. IPv6 multicast addresses begin with ff00::/8, with examples including ff02::1 for all IPv6 nodes on the local link and ff02::2 for all IPv6 routers on the local link.
5. Unlike IPv4, IPv6 does not use broadcast addressing; functions that previously relied on broadcasts are commonly implemented using multicast and Neighbor Discovery.

Lecture reference: Section 4.2 – IPv6 Multicast and Anycast, Slides 61–62.

Question 20. IPv6 Autoconfiguration Using SLAAC and DHCPv6

An industrial computer is connected to an IPv6-enabled Ethernet network for the first time. The device must automatically establish communication without the engineer manually entering a complete IPv6 address and router configuration. Explain the IPv6 autoconfiguration process, including link-local addressing, Duplicate Address Detection (DAD), Router Solicitation (RS), Router Advertisement (RA), SLAAC, and DHCPv6. Discuss how the computer learns its network configuration.

Good Answer:

1. When the industrial computer starts, it creates an IPv6 link-local address that allows communication with other devices on the same network link.
2. The computer performs Duplicate Address Detection (DAD) to verify that its proposed IPv6 address is not already being used by another device.
3. The computer can send a Router Solicitation message to request information, and a router responds with a Router Advertisement containing available network prefixes and router information.
4. If Stateless Address Autoconfiguration (SLAAC) is permitted, the computer generates its own IPv6 address using the advertised network prefix and an interface identifier.
5. DHCPv6 may provide addresses and additional configuration parameters such as DNS information, while Router Advertisements help hosts determine the available routers, prefixes, and appropriate autoconfiguration method.

Lecture reference: Section 4.2 – IPv6 Autoconfiguration: SLAAC, DHCPv6, and Host Startup, Slides 63–64.